A force diagram begins with one chosen object

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

External force / ES: fuerza externa / PT: força externa

A force from outside the chosen boundary.

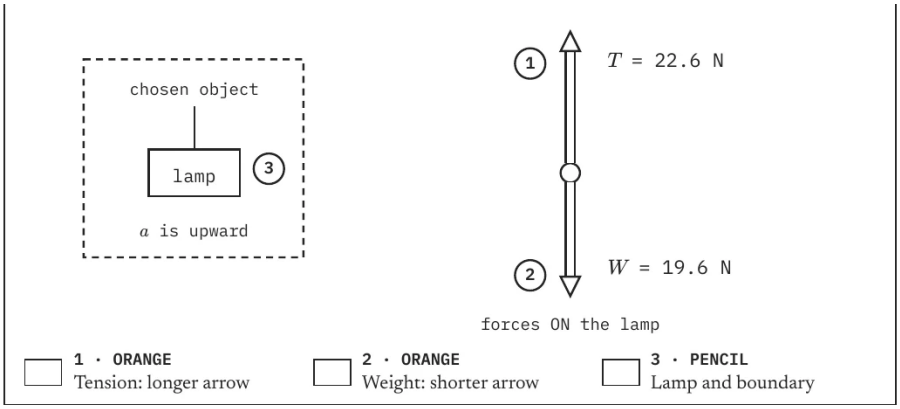
Also on your note page: Object, Force-arrow scale, Net force.

B READ THE RELATIONSHIP

T − mg = ma for the lamp, with up positive

As you read, find: object · boundary · outside.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A rope lifts a 2 kg lamp with upward acceleration 1.5 m/s² . Find the tension. Use g = 9.8 m/s² .

Given m = 2 kg a y = +1.5 m/s² g = 9.8 m/s²

Rule Up is +. T − mg = ma y .

Work W = 2(9.8) = 19.6 N T = 19.6 + 2(1.5)

Answer T = 22.6 N T and W are not a third-law pair.

E SAVE FOR THE FINAL MEETING

An XJ force diagram contains forces on XJ, not the forces XJ exerts on other bodies. Defining that boundary keeps the acceleration calculation consistent.

Read the physics: External force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Draw a dot or box for the object. Name each outside object that pushes or pulls it. Draw one arrow for each force on your chosen object.

02 HOW IT WORKS

Label each arrow with the source or force name. Longer arrows mean stronger forces when the scale is the same. Put velocity outside the force diagram.

03 WHAT TO CHECK

A hanging lamp has gravity from Earth and tension from the rope. Equal forces balance on the lamp. They are not a third-law pair: both act on the lamp.

04 TRY THE EXAMPLE FROM YOUR NOTES

A rope lifts a 2 kg lamp with upward acceleration 1.5 m/s² . Find the tension. Use g = 9.8 m/s² . T = 22.6 N T and W are not a third-law pair.

Model check: T − mg = ma for the lamp, with up positive.

05 CONNECT TO XJ

An XJ force diagram contains forces on XJ, not the forces XJ exerts on other bodies. Defining that boundary keeps the acceleration calculation consistent.

SAY IT BEFORE YOU WRITE

Which object receives every force in this diagram? I drew ___ because ___ acts on the chosen object.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Object

ES: objeto / PT: objeto

Meaning in my words	How can you tell?
In a force diagram, what do we mean by the object? In my words, this means ___.	You circle only the cart. Which force belongs: the hand on the cart or the cart on the hand? I can tell because ___.
Example or sketch	**Non-example and why**
What would you put at the centre of a free-body diagram for a hanging lamp? My example shows ___.	A student switches to forces on the ceiling halfway through. How should they organize those? I would change ___ because ___.

External force

ES: fuerza externa / PT: força externa

Meaning in my words	How can you tell?
How would you explain external force to a friend? In my words, this means ___.	Draw a boundary around a lamp alone. Why is the cable's pull external? I can tell because ___.
Example or sketch	**Non-example and why**
If your system contains two carts and a hand pushes one from outside, what force would you include? My example shows ___.	If lamp and cable are both inside the boundary, should their mutual pull count as an external force? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force-arrow scale

ES: escala de las flechas de fuerza / PT: escala das setas de força

Meaning in my words	How can you tell?
How would you explain force-arrow scale to a friend? In my words, this means ___.	Your scale is 1 cm for 2 N. How long should a 6 N arrow be? I can tell because ___.
Example or sketch	**Non-example and why**
Draw 2 N left and 6 N right at that scale. My example shows ___.	Two arrows of equal length are labeled 2 N and 8 N on one diagram. What would you change? I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the lamp, its boundary, and its two force arrows. Make their lengths predict upward acceleration.

Labels you can use: object · boundary · outside.

Chosen object: _____



Tell a partner

I drew ___ because ___ acts on the chosen object.

Build the force overlay

15

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Keep only forces on the chosen object in its diagram.

CHOOSE ONE JOB TOGETHER

SYSTEMS ENGINEER ☐

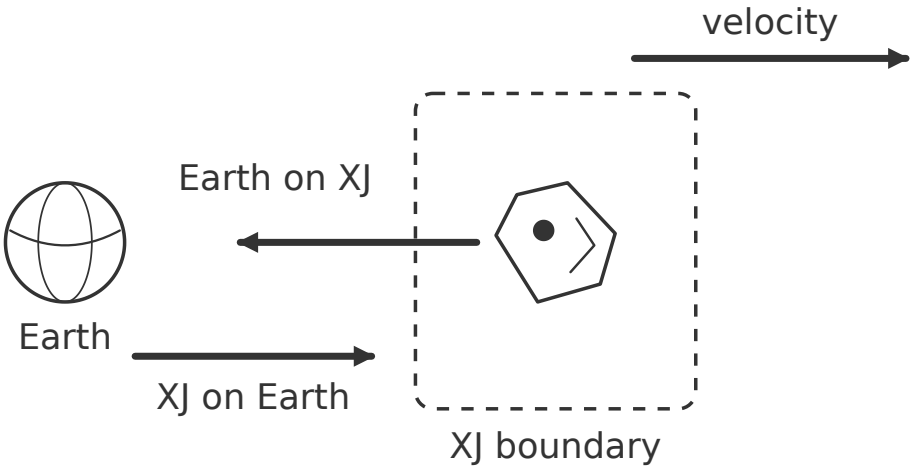
Outline XJ as the chosen object. Keep only the force arrows you would use to calculate XJ's acceleration.

ANIMATION DIRECTOR ☐

Separate the force and motion layers. Tell the animator which Earth arrow belongs in a different shot.

Word help: boundary / force / motion

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.



01 OUR FIRST CALL

Which arrows belong in 'forces on XJ'? Explain what you keep and move.

This arrow acts on ___, so it belongs ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Where does the new arrow belong? Update the force layer and explain.

Now we would say ___ because ___.

Switch speakers. Must XJ move in the direction of its net force? Give a possible motion that shows your reasoning.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks